Unified Language Model Pre-training for Natural Language Understanding and Generation

Dong et al. (NeurIPS 2019), arXiv:1905.03197

Paper summary for the CS-781 capstone

The paper at a glance

Citation

L. Dong, N. Yang, W. Wang, F. Wei, X. Liu, Y. Wang, J. Gao, M. Zhou, H.-W. Hon. *Unified Language Model Pre-training for Natural Language Understanding and Generation*. NeurIPS 2019. Microsoft Research.

One-sentence summary

One shared Transformer is pre-trained as a bidirectional, a unidirectional *and* a sequence-to-sequence language model, simply by switching the **self-attention mask**.

Why it matters for our capstone

Our A3 prefix-LM uses UniLM's sequence-to-sequence mask: bidirectional over the prompt, causal over the continuation, in one stack.

Where it appears in our slides

Main deck: "A3: hybrid prefix-LM with ALiBi" and "Attention masks".

Roadmap

1. The problem
2. The basic idea: one network, many masks
3. Results
4. Assessment and outlook

Understanding models vs generation models

Objective	ELMo	GPT	BERT	UniLM
Left-to-right LM	✓	✓		✓
Right-to-left LM	✓			✓
Bidirectional LM			✓	✓
Seq-to-seq LM				✓

- **BERT** (bidirectional) is excellent for understanding tasks (NLU), but its bidirectionality makes generation (NLG) hard.
- **GPT** (left-to-right) generates naturally but sees only left context when reading.

The question

Can *one* pre-trained network do both jobs: read a legal document with full context, *and* write a summary of it?

Claimed advantages of unifying

1. One model to train and host, not several.
2. Shared parameters learn more general representations and overfit less to any one objective.
3. The seq-to-seq mode suits summarisation.

The mask decides what context a token sees

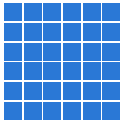
Standard Transformer self-attention, plus a mask matrix M:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V$$

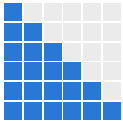
$$M_{ij} = \begin{cases} 0 & \text{token } i \text{ may attend to } j \\ -\infty & \text{blocked} \end{cases}$$

Same weights, same architecture. Only M changes between objectives.

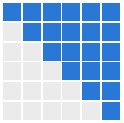
Segment embeddings tell the model which objective it is being trained on.



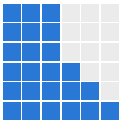
bidirectional (like BERT)



left-to-right (like GPT)



right-to-left



seq-to-seq (source = 3 tokens)

Rows = queries, columns = keys; blue = allowed.

The sequence-to-sequence mask, step by step

Input: `[SOS] S1 [EOS] S2 [EOS]`, with source S1 and target S2.

- **Source tokens** see all source tokens, in both directions, but no target tokens.
- **Target tokens** see all source tokens, plus target tokens to their left.

Paper's example: source $t_1 t_2$, target $t_3 t_4 t_5$: `[SOS] t1 t2 [EOS] t3 t4 t5 [EOS]`. Both t_1 and t_2 see the first four tokens. t_4 sees the first six.

The encoder and decoder are the *same* Transformer. There is no cross-attention.

Easy legal example

Source: "*appeal against conviction under Section 302; evidence found unreliable*". Target: "*appeal allowed*".

- "unreliable" can look *forward* to "evidence" and "found" while reading, which is impossible in GPT.
- "allowed" can look at the whole source and at "appeal" in the target, but not at later target words.

Reading is bidirectional; writing is causal.

Pre-training objectives: all cloze tasks

Every objective is a **cloze** task: replace random word pieces with `[MASK]` and predict the originals. Only the mask decides which context is visible.

Mixing within each batch:

- 1/3 bidirectional LM (plus next-sentence prediction, as in BERT);
- 1/3 sequence-to-sequence LM;
- 1/6 left-to-right LM and 1/6 right-to-left LM.

Masking: 15% of tokens; of these 80% become `[MASK]`, 10% a random token, 10% unchanged. 80% of the time one token is masked, 20% of the time a bigram or trigram.

Easy example: left-to-right cloze

"x_1 x_2 `[MASK]` x_4": with the left-to-right mask, the prediction for `[MASK]` can use only x_1, x_2 and itself.

With the bidirectional mask, the same slot can also use x_4.

Difference from our A3

UniLM predicts *masked* tokens. Our A3 predicts every *next* token after the prefix, with no `[MASK]` tokens. The attention mask is the same idea.

Model and training setup

- Architecture of BERT-large: 24 layers, hidden size 1,024, 16 heads, about **340M parameters**; GELU; tied output embeddings.
- **Initialised from BERT-large**, then pre-trained on English Wikipedia and BookCorpus.
- WordPiece vocabulary of 28,996; maximum length 512.
- Adam, learning rate 3×10^{-5}, 40k warm-up steps, linear decay; dropout 0.1; weight decay 0.01; batch 330.
- About 770,000 steps. Roughly 7 hours per 10,000 steps on 8 V100 GPUs with mixed precision.

Fine-tuning

- **NLU** (e.g. classification): use the bidirectional mask; classify from the `[SOS]` vector, like BERT's `[CLS]`.
- **NLG** (e.g. summarisation): pack `[SOS]` `document` `[EOS]` `summary` `[EOS]`, use the seq-to-seq mask, and mask target tokens to recover.
- Masking the final `[EOS]` teaches the model when to *stop*.

Abstractive summarisation

CNN/DailyMail (ROUGE F1):

Model	R-1	R-2	R-L
LEAD-3 (first 3 sentences)	40.42	17.62	36.67
Best extractive	43.25	20.24	39.63
PGNet	39.53	17.28	37.98
Bottom-Up	41.22	18.68	38.34
UniLM	**43.33**	20.21	**40.51**

Gigaword (ROUGE F1):

Model	R-1	R-2	R-L
only 10K training examples			
Transformer	10.97	2.23	10.42
MASS	25.03	9.48	23.48
UniLM	**32.96**	**14.68**	**30.56**
full 3.8M examples			
MASS	37.66	18.53	34.89
UniLM	**38.45**	**19.45**	**35.75**

New state of the art for abstractive summarisation. With only 10K examples UniLM beats MASS by 7.08 ROUGE-L, a strong **low-resource** result. Inputs were truncated to 640 tokens (CNN/DM); beam 5 with duplicate-trigram blocking.

Question answering and question generation

Task	Model	Score
SQuAD 2.0 (extractive, F1)	BERT-large	81.8
	UniLM	**83.4**
CoQA (extractive, F1)	BERT-large	82.7
	UniLM	**84.9**
CoQA (*generative*, F1)	PGNet	45.4
	UniLM	**82.5**
SQuAD question gen. (BLEU-4)	SemQG	18.37
	UniLM	**22.12**

Highlights

- Generative QA jumps by **37.1 F1**, nearly closing the gap to extractive QA.
- Using UniLM to *generate* 5M answerable and 4M unanswerable training questions raised SQuAD 2.0 F1 from 83.4 to **87.6**.
- Dialogue response generation (DSTC7): NIST-4 of 2.67, above the reported human score of 2.65.

Language understanding (GLUE)

GLUE test set:

Model	CoLA	SST-2	MNLI-m	RTE	Score
GPT	45.4	91.3	82.1	56.0	72.8
BERT-large	60.5	94.9	86.7	70.1	80.5
UniLM	**61.1**	94.5	**87.0**	**70.9**	**80.8**

Reading it

UniLM matches BERT-large on understanding (80.8 vs 80.5), while *also* being a strong generator. Adding unidirectional and seq-to-seq objectives did **not** hurt understanding.

Connection to our A3 prefix-LM

	UniLM	Our A3
Stack	one shared Transformer	one shared Transformer
Key mask	sequence-to-sequence (source bidirectional)	prefix: prompt bidirectional, continuation causal
Objective	cloze: predict `[MASK]` tokens	next-token prediction after the prefix
Mixing	1/3 bi, 1/3 seq2seq, 1/3 uni	50% plain causal, 50% random prefix
Positions	learned (from BERT)	ALiBi, no position embeddings
Start point	initialised from BERT-large	trained from scratch (rules)
Size	about 340M	81.3M

What UniLM suggests we test

Does bidirectional reading of the 512-word prompt help A3 beat the causal A1 on continuation loss and BERTScore? Is 50% causal mixing the right ratio?

Strengths and limitations

Strengths

- One model for understanding *and* generation.
- Very simple mechanism: only the attention mask changes.
- Strong generation results, especially with little data.
- No loss on understanding benchmarks compared with BERT.
- Parameter sharing between "encoder" and "decoder" halves the size of an equivalent encoder–decoder.

Limitations

- Initialised from BERT-large, so gains are not purely from the unified objective.
- No ablation isolating each objective's contribution; the authors list this as future work.
- Cloze objectives train on only about 15% of tokens per pass, which is less efficient than predicting every next token.
- Maximum length 512 limits long-document tasks.
- Encoder and decoder must share one set of weights and one sequence budget.

Future research directions (1): from the paper and after

Direction	Details
Scale up *(paper)*	More epochs, larger models, web-scale corpora
Ablations *(paper)*	Measure the benefit of pre-training several LM tasks in one network
Cross-lingual UniLM *(paper)*	Extend beyond monolingual tasks
Multi-task fine-tuning *(paper)*	Fine-tune on NLU and NLG tasks together, as in MT-DNN
Better unified objectives	UniLMv2 (Bao et al., 2020) uses pseudo-masked LM to train both modes efficiently
Blank infilling	GLM (Du et al., 2022) generates masked spans autoregressively with 2D positions
Mixture of denoisers	UL2 (Tay et al., 2023) mixes prefix-LM and span-corruption objectives
Systematic comparison	T5 (Raffel et al., 2020) compares prefix-LM, decoder-only and encoder–decoder models directly

Future research directions (2): open questions

Open in general

- What mixture of objectives is best for a fixed compute budget?
- Do unified models scale as well as pure decoder-only models?
- How to reuse a bidirectionally-read prompt in efficient KV-cached generation.

Open for legal NLP

- One in-house model that both classifies (Phases 3–5) and writes (Phases 1, 6), which suits a law firm's hosting limits.
- Using a unified model to generate training questions for Phase 8 (QA), as UniLM did for SQuAD.
- Long judgments: combining the prefix mask with local attention.

Summary

- BERT reads well; GPT writes well. UniLM does both with **one shared Transformer**.
- The **self-attention mask** selects the mode: bidirectional, left-to-right, right-to-left, or sequence-to-sequence (source bidirectional, target causal).
- All modes are trained as cloze tasks, mixed in every batch.
- Results: new state of the art on five generation tasks (e.g. CNN/DM ROUGE-L 40.51, CoQA generative F1 82.5) while matching BERT on GLUE (80.8).
- For us: the idea behind the A3 prefix-LM mask.

Reference

Dong, L., Yang, N., Wang, W., Wei, F., Liu, X., Wang, Y., Gao, J., Zhou, M., and Hon, H.-W. (2019). Unified Language Model Pre-training for Natural Language Understanding and Generation. In *Advances in Neural Information Processing Systems 32*. arXiv:1905.03197.

Later work cited: Bao et al. (2020) UniLMv2; Raffel et al. (2020) T5; Du et al. (2022) GLM; Tay et al. (2023) UL2.